
Data-Free Bayesian Test-Time Model Merging with Soft Batch Normalization Buffer Fusion

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Abstract

Test-Time Model Merging (TTMM) is an emerging paradigm for dynamically fusing multiple task-specific expert neural networks into a single multi-task model at deployment time. It adapts to a non-stationary stream of test data without parameter backpropagation. However, existing TTMM methods face three fundamental challenges: (1) a critical reliance on private source calibration data for parameter-sensitivity (Fisher Information) estimation, (2) severe activation mismatches due to the omission or hard-switching of Batch Normalization (BN) running statistics, and (3) a susceptibility to the “feedback trap” (a failure mode of unconstrained entropy minimization where parameters irreversibly collapse onto a single confidently wrong expert, which is particularly severe when experts share a classification head). To resolve these limitations, we propose DF-Bayes-TTMM (Data-Free Bayesian Test-Time Model Merging with Soft BN Buffer Fusion). First, we formulate TTMM as a dynamic Bayesian mixture-of-experts, estimating soft, continuous posterior probabilities for each active expert on-the-fly based on predictive confidence. Second, we introduce Soft BN Buffer Fusion, which continuously blends expert running statistics using the posterior weights to preserve activation integrity without hard-switching discontinuities. We mathematically prove that our fusion equations perfectly reconstruct the moments of the underlying mixture-of-Gaussians activation distribution. Third, we establish a fully data-free, uncertainty-guided novelty detection mechanism based on the average ex-

pert entropy. Our extensive empirical evaluations on non-stationary vision streams show that DF-Bayes-TTMM completely eliminates private source data dependency, avoids the feedback trap, and achieves state-of-the-art performance—enhancing robustness and stability under open-world shifts. In doing so, it overcomes the catastrophic collapse of unconstrained entropy minimization (improving accuracy by up to 28%) and outperforms competitive hard-routing baselines in open-world settings while maintaining a continuous and principled design.

1. Introduction

The paradigm of pre-training large foundation models and fine-tuning them on specialized downstream domains has led to a proliferation of specialized “expert” models (Wortsman et al., 2022; Radford et al., 2021). While each expert excels on its specific task, deploying multiple independent experts for multi-task applications is computationally prohibitive. To integrate disparate expert capabilities at inference time, model merging techniques such as task arithmetic (Ilharco et al., 2023), Ties-Merging (Yadav et al., 2023), and Fisher weighted averaging (Matena & Raffel, 2022) have been developed. These methods interpolate the weights of specialized models in parameter space to yield a single unified model without joint training.

To handle dynamic shifts, Test-Time Model Merging (TTMM) adapts merging coefficients on-the-fly via unsupervised objectives (Yang et al., 2023; 2024). By tuning low-dimensional coefficients instead of full parameters, TTMM offers a highly efficient alternative to standard Test-Time Adaptation (TTA) (Wang et al., 2021), preventing representational collapse and forgetting with minimal compute.

Despite their potential, existing TTMM frameworks are restricted by three fundamental bottlenecks:

1. Source Data Inaccessibility: Existing methods pre-compute Fisher Information on source

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

datasets to scale learning rates. However, training data is often private or inaccessible. Fully data-free preconditioning is essential.

2. Activation Mismatch: Existing TTMM focus solely on weight interpolation, omitting Batch Normalization (BN) statistics. This causes severe activation mismatches. DR-Fisher’s hard switching introduces discontinuities.
3. The Feedback Trap: Unconstrained entropy minimization can cause coefficients to collapse toward a single out-of-distribution (OOD) expert that is spuriously confident, a failure mode that is particularly severe when experts share a classification head.

To overcome these boundaries, we introduce DF-Bayes-TTMM (Data-Free Bayesian Test-Time Model Merging with Soft BN Buffer Fusion), a mathematically rigorous, fully unsupervised, and completely data-free TTMM framework.

We formulate TTMM as a dynamic Bayesian mixture-of-experts. Rather than performing hard routing, we estimate the continuous posterior probabilities of the experts on-the-fly using their predictive confidence on the incoming test batch. This posterior distribution provides a principled, soft-routing weight vector that guides both parameter and buffer fusion.

To resolve activation mismatches, we propose Soft BN Buffer Fusion, dynamically blending expert running statistics using posterior weights. This blends moments of the underlying mixture-of-Gaussians distribution, preserving activation integrity without discontinuities.

Furthermore, we design an uncertainty-guided novelty detection mechanism based on the average expert entropy (which serves as a robust indicator of epistemic uncertainty). Our model robustly identifies novel domains in the open-world stream without relying on private source prototypes or distance thresholds.

Finally, we introduce a Maximum A Posteriori (MAP) adaptation step preconditioned by Test-Time Fisher (TT-Fisher) sensitivities estimated on test samples. A temporal Gaussian prior regularizes coefficient updates, ensuring stable, data-free adaptation.

We evaluate DF-Bayes-TTMM on non-stationary streams comprising MNIST, KMNIST, and FashionMNIST. Our results show that DF-Bayes-TTMM completely avoids the feedback trap, achieves perfect novelty routing, and delivers state-of-the-art performance, enhancing robustness while ensuring continuous, data-free adaptation.

2. Related Work

Model Merging: Model merging aims to combine multiple neural networks trained on different tasks or domains into a single unified network without the cost of retraining (Wortsman et al., 2022; Ainsworth et al., 2023; Jordan et al., 2022). Standard weight averaging, however, suffers from parameter interference. To mitigate this, Task Arithmetic (Ilharco et al., 2023), Ties-Merging (Yadav et al., 2023), and DARE (Yu et al., 2023) sparsify and scale weight deltas. Fisher Merging (Matena & Raffel, 2022) uses diagonal Fisher Information matrices as parameter-wise weights. RegMean (Jin et al., 2023) solves a quadratic optimization problem to minimize activation distance. BECAME (Li et al., 2025) proposes Bayesian continual learning to compute closed-form merging coefficients. However, these methods are static and require offline calibration data.

Test-Time Adaptation (TTA): TTA adapts a pre-trained model to target domain shifts on-the-fly using unlabeled test streams (Liang et al., 2023; Sun et al., 2020). The foundational baseline Tent (Wang et al., 2021) optimizes scale and shift parameters of BN layers by minimizing prediction entropy. Under continuous, non-i.i.d. shifts, TENT suffers from catastrophic parameter drift and representational collapse. Advanced TTA methods use weight regularization (Niu et al., 2022), mean teacher frameworks (Wang et al., 2022), and robust statistics estimation (Yuan et al., 2023). While effective, full TTA is computationally expensive and prone to collapse under severe shifts.

Test-Time Model Merging (TTMM): TTMM combines the parameter efficiency of model merging with the dynamic flexibility of TTA by adapting merging coefficients online (Yang et al., 2024; Zhao et al., 2024). AdaMerging (Yang et al., 2023; 2024) adaptively learns task-wise or layer-wise merging coefficients by minimizing prediction entropy with a uniform learning rate, which often leads to representational collapse in sensitive layers. FP-CA (Luan et al., 2026) and IGGS-Merge (Authors, 2026a) use pre-computed Fisher Information to precondition and damp updates in sensitive layers. Recently, PROTO-TTMM (Authors, 2026b) addresses open-world settings using prototype cohesion and contrastive alignment. Our work directly addresses their limitations: they require private source data, completely omit BN statistics, and collapse due to the feedback trap on shared classification heads.

3. Methodology

3.1. Problem Formulation

Let θ_{base} denote a shared base pre-trained model, and let $\{\theta_k\}_{k=1}^K$ be a library of K expert networks fine-tuned from θ_{base} on K distinct domains. The task vector for expert k is defined as $v_k = \theta_k - \theta_{\text{base}}$. At test time, the model receives a continuous, non-stationary stream of unlabeled batches $X_1, X_2, \dots, X_t$ from different domains. Our goal is to dynamically merge the expert parameters θ_k into a single merged model $\theta_{\text{merged},t}$ at each step t using layer-wise merging coefficients $\Lambda_{l,t}$ (for each layer l) and a global coefficient vector $\lambda_{\text{global},t}$ (collectively denoted as $\Lambda_t = \{\Lambda_{l,t}, \lambda_{\text{global},t}\}$) to maximize classification accuracy across the stream. Specifically, for each layer l of the model, the merged parameters are computed as:

$$\theta_{\text{merged},t,l}(\Lambda_{l,t}) = \theta_{\text{base},l} + \sum_{k=1}^K \lambda_{l,t,k} v_{l,k} \quad (1)$$

where $\theta_{\text{merged},t,l}$ and $\theta_{\text{base},l}$ denote the parameters of layer l , $v_{l,k} = \theta_{l,k} - \theta_{\text{base},l}$ is the task vector of expert k at layer l , and $\lambda_{l,t,k} \geq 0$ is the merging coefficient for expert k at layer l under the simplex constraint $\sum_k \lambda_{l,t,k} = 1$ for all layers. In addition to these layer-wise coefficients, we maintain a global coefficient vector $\lambda_{\text{global},t}$ which is used to dynamically fuse the Batch Normalization statistics during test-time adaptation, as detailed in Sections 3.3 and 3.4.

3.2. Continuous Bayesian Mixture of Experts

To estimate the contribution of each expert to the current test batch X , we formulate TTMM as a dynamic Bayesian mixture-of-experts. We define the unsupervised log-likelihood of the batch X under expert k as the negative Shannon entropy of the expert's predictions:

$$\log P(X \mid \text{expert } k) = -\gamma H(Y \mid X, \theta_k) \quad (2)$$

where $\gamma > 0$ is a temperature scaling parameter, and $H(Y \mid X, \theta_k) = -\frac{1}{|X|} \sum_{i \in X} \sum_c p_{i,c} \log p_{i,c}$ is the average prediction entropy on the batch. Experts that are highly confident on the test batch will yield low entropy, translating to high likelihood.

Using Bayes' theorem, we compute the posterior probability of each expert k being the active expert for the current batch:

$$w_k = P(\text{expert } k \mid X) = \frac{L_k P(\text{expert } k)}{\sum_{j=1}^K L_j P(\text{expert } j)} \quad (3)$$

where $L_k = \exp(-\gamma H(Y \mid X, \theta_k))$ represents the likelihood under expert k , and $P(\text{expert } k)$ is a uniform

prior over experts. This posterior probability vector $w = [w_1, \dots, w_K]$ provides a continuous, soft-routing weight distribution over the expert library.

3.3. Soft Batch Normalization Buffer Fusion

Omitting Batch Normalization statistics during weight fusion leads to severe activation mismatches. Hard-routing methods select the running statistics of a single active expert, which creates sharp discontinuities at task boundaries. To resolve this, we propose Soft BN Buffer Fusion, which continuously blends expert running statistics.

To derive the fusion equations, we make a standard simplifying assumption about the intermediate representation space:

Assumption 3.1. The activation distribution for any given feature channel in a Batch Normalization layer, conditioned on a specific expert k , is modeled as a Gaussian distribution: $z \mid \text{expert } k \sim \mathcal{N}(\mu_k, \sigma_k^2)$.

Under this assumption, we model the total activation distribution under our mixture-of-experts as a mixture of Gaussians.

Proposition 3.2. Let z represent an activation variable. If Assumption 1 holds and the active expert is selected according to the posterior probabilities $w_k = P(\text{expert } k \mid X)$, then the combined activation z is distributed as a mixture of Gaussians $z \sim \sum_{k=1}^K w_k \mathcal{N}(\mu_k, \sigma_k^2)$. The mean μ_{fused} and variance σ_{fused}^2 of this mixture are given exactly by:

$$\mu_{\text{fused}} = \sum_{k=1}^K w_k \mu_k \quad (4)$$

$$\sigma_{\text{fused}}^2 = \sum_{k=1}^K w_k (\sigma_k^2 + (\mu_k - \mu_{\text{fused}})^2) \quad (5)$$

Proof. By the law of total expectation, the mean of the mixture activation is:

$$\begin{aligned} \mu_{\text{fused}} &= \mathbb{E}[z] = \sum_k P(\text{expert } k \mid X) \mathbb{E}[z \mid \text{expert } k] \\ &= \sum_{k=1}^K w_k \mu_k \end{aligned} \quad (6)$$

Using the variance decomposition formula (law of total variance):

$$\text{Var}(z) = \mathbb{E}[\text{Var}(z \mid \text{expert } k)] + \text{Var}(\mathbb{E}[z \mid \text{expert } k]) \quad (7)$$

The first term is the expected conditional variance:

$$\mathbb{E}[\text{Var}(z \mid \text{expert } k)] = \sum_{k=1}^K w_k \sigma_k^2 \quad (8)$$

The second term is the variance of the conditional expectation:

$$\text{Var}(\mathbb{E}[z \mid \text{expert } k]) = \sum_{k=1}^K w_k (\mu_k - \mu_{\text{fused}})^2 \quad (9)$$

Combining these yields the exact variance of the mixture activation:

$$\sigma_{\text{fused}}^2 = \sum_{k=1}^K w_k \sigma_k^2 + \sum_{k=1}^K w_k (\mu_k - \mu_{\text{fused}})^2 \quad (10)$$

which matches the proposed equation. $\square$

By computing the exact first and second moments of the mixture activation distribution, the Batch Normalization layers can then utilize these exact moments to correctly normalize the incoming activations on-the-fly. This serves as a mathematically sound and principled approximation for normalizing the full (non-Gaussian) mixture activation profile, preserving representational integrity across arbitrary domain combinations, eliminating activation mismatches, and avoiding hard-switching discontinuities.

3.4. Maximum A Posteriori Layer-Wise Adaptation

To understand the dual role of the Batch Normalization (BN) fusion weights, we distinguish between the initial probabilistic routing and the inner-loop fine-tuning. We formulate our adaptation as a dual-phase hierarchy: the Bayesian posterior initialization acts as the primary adaptation driver, offering extremely high-precision routing on-the-fly, while the subsequent inner-loop MAP optimization serves as a secondary, robust insurance policy. Upon receiving a test batch X_t , we first compute the expert posterior probabilities w (Eq. 5), representing our probabilistic domain routing recommendation. We use this posterior w to initialize the layer-wise coefficients $\lambda_{l,k}$ and the global coefficient vector $\lambda_{\text{global},k}$ in a logit-space, such that their initial softmax probabilities align with the Bayesian posterior:

$$\lambda_{l,k} \propto \log(w_k + \epsilon), \quad \lambda_{\text{global},k} \propto \log(w_k + \epsilon) \quad (11)$$

where $\epsilon = 10^{-10}$ prevents log-zero errors. Crucially, the initial Soft BN Buffer Fusion (Eq. 7–8) is applied using this initial posterior weight vector w . This places the merged model in an optimal starting configuration.

During the subsequent inner MAP adaptation loop, the layer-wise merging coefficients $\lambda_{l,k}$ are updated to minimize the predictive entropy of the merged model. As these parameters adapt, the representations and activation statistics change. To ensure

that the BN running statistics remain perfectly synchronized with these parameter updates, the global coefficient vector λ_{global} is optimized alongside the layer coefficients. During the M inner steps, we dynamically update the BN running statistics using the current softmax-transformed global coefficients, i.e., $w_{\text{inner}} = \text{softmax}(\lambda_{\text{global}})$. This dual-phase mechanism maintains statistical and parameter synchronization throughout the adaptation process. The motivation for maintaining separate layer-wise and global coefficients lies in their distinct roles: while the layer-wise coefficients resolve local “representational friction” between disparate expert layers, a single, global coefficient vector provides a more stable and holistic estimate of the domain mixture for normalizing activations. Constraining the BN fusion to a single global vector avoids over-parameterization and prevents local statistical distortions. Under a temporal Gaussian prior centered at the previous step’s coefficients Λ_{t-1} , the objective to minimize is:

$$\mathcal{L} = H(Y \mid X, \theta_{\text{merged}}) + \frac{\beta}{2} \|\Lambda_t - \Lambda_{t-1}\|^2 \quad (12)$$

where $\beta \geq 0$ represents the prior regularization strength. In practice, all layer-wise merging coefficients $\lambda_{l,k}$ and the global coefficient vector $\lambda_{\text{global},k}$ are grouped together as Λ_t , and they are optimized jointly using a single Adam optimizer step in the inner loop. This temporal prior stabilizes adaptation for both parameters and statistics, preventing catastrophic drift and coefficient oscillation.

To adapt layer-wise coefficients safely under heterogeneous sensitivities, we apply Test-Time Fisher (TT-Fisher) preconditioning. We estimate parameter-level diagonal sensitivities dynamically on-the-fly using the current unlabeled test batch X and the model’s own predictions as pseudo-labels:

$$F(p) = \frac{1}{|X|} \sum_{i,c} \hat{p}_{i,c} [\nabla_p \log P(y_i = c \mid x_i, \theta_{\text{merged}})]^2 \quad (13)$$

We then compute the layer-wise sensitivity score $S_l = \sum_{p \in \text{layer } l} F(p)$. During backpropagation, the gradients of the layer-wise merging coefficients are scaled inversely to their layer sensitivities:

$$\nabla_{\lambda_l} \mathcal{L} \leftarrow \nabla_{\lambda_l} \mathcal{L} \cdot \frac{1}{S_l + \epsilon_d} \quad (14)$$

This data-free preconditioning dampens adaptation in highly sensitive layers (e.g., classifier heads and early convolutions) and accelerates stable adaptation in robust intermediate representation blocks.

While highly effective, we note that the TT-Fisher (Eq. 12) is an empirical Fisher Information matrix es-

timated on a single test batch using the model’s own predictions as pseudo-labels. Consequently, it represents a noisy, local approximation of the true Fisher Information. In scenarios with extremely small batch sizes B or high predictive uncertainty (e.g., when transitioning between domains or encountering a novel domain), this estimator can experience minor instability. However, our empirical results in Section 5.3.6 demonstrate that DF-Bayes-TTMM is highly robust across a wide range of batch sizes $B \in \{16, 32, 64, 128\}$, indicating that the inverse sensitivity scaling is sufficiently robust to tolerate these localized approximation noises in practice.

3.5. Uncertainty-Guided Novelty Detection

In an open-world setting, a completely novel domain D_{novel} may arrive. Since there is no pre-trained expert for the novel domain, all experts will exhibit high predictive entropy. We compute the epistemic uncertainty of the expert mixture via the Mutual Information (MI) between the predictions and the expert identity:

$$I(Y; \text{Expert} \mid X) = H(Y \mid X) - \sum_{k=1}^K w_k H(Y \mid X, \theta_k) \quad (15)$$

where $H(Y \mid X) = -\sum_c \bar{p}_c \log \bar{p}_c$ is the entropy of the average prediction $\bar{p} = \sum_k w_k p_k$.

While mutual information provides the theoretical grounding for quantifying epistemic uncertainty under confident disagreement, it can become uninformative when all experts are equally unconfident. Specifically, when a completely novel domain arrives, every expert’s predictions approach a uniform distribution, yielding high individual predictive entropies $H(Y \mid X, \theta_k) \approx H_{\text{max}}$ (aleatoric uncertainty). Consequently, the posterior probabilities w_k also approach a uniform distribution $1/K$, causing the average prediction entropy $H(Y \mid X)$ to also reach H_{max} . Under this uniform consensus, both terms in the MI equation approach H_{max} , resulting in a low MI score ($I(Y; \text{Expert} \mid X) \approx 0$) that fails to flag the novel domain.

In contrast, the average expert entropy $\bar{H} = \frac{1}{K} \sum_{k=1}^K H(Y \mid X, \theta_k)$ remains consistently and maximally high ($\approx H_{\text{max}}$), acting as a monotonic and highly discriminative indicator of the absolute lack of domain-specific expertise. Thus, the average expert entropy serves as an extremely robust, direct, and computationally efficient proxy for detecting novel domains on-the-fly. We therefore define our novelty detection rule as:

$$\text{is_novel} = \mathbb{I}(\bar{H} > \tau_N) \quad (16)$$

If detected as novel, we initialize the merging coeffi-

cients uniformly and adapt them to find a parameter configuration that generalizes to the novel domain, while bypassing the feedback trap by setting the BN running statistics to a uniform average to prevent representational decay.

The complete step-by-step process of DF-Bayes-TTMM is summarized in Algorithm 1.

Algorithm 1 DF-Bayes-TTMM: Data-Free Bayesian Test-Time Model Merging

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1: Input: Expert models  $\{\theta_k\}_{k=1}^K$ , pre-trained running
   statistics  $\{\mu_k, \sigma_k^2\}_{k=1}^K$ , test stream  $\{X_t\}_{t=1}^T$ , parameters
    $\gamma, \beta, \tau_N$ , inner steps  $M = 3$ .
2: Initialize:  $\Lambda_0$  uniformly,  $\Lambda_{\text{prev}} \leftarrow \Lambda_0$ 
3: for each test batch  $X_t$  in stream do
4:   Evaluate expert entropies  $H_k = H(Y \mid X_t, \theta_k)$  for
      $k = 1, \dots, K$ 
5:   Compute average expert entropy  $\bar{H} = \frac{1}{K} \sum_k H_k$ 
6:   if  $\bar{H} > \tau_N$  then
7:     is_novel  $\leftarrow$  True
8:     Initialize layer and global coefficients:  $\lambda_{l,k} \leftarrow 0$ ,
        $\lambda_{\text{global},k} \leftarrow 0$ 
9:     Set BN fusion weights  $w \leftarrow [1/K, \dots, 1/K]$ 
10:   else
11:     is_novel  $\leftarrow$  False
12:     Compute posterior  $w_k = \frac{\exp(-\gamma H_k)}{\sum_j \exp(-\gamma H_j)}$ 
13:     Initialize layer and global coefficients:  $\lambda_{l,k} \leftarrow$ 
        $\log(w_k + \epsilon)$ ,  $\lambda_{\text{global},k} \leftarrow \log(w_k + \epsilon)$ 
14:   end if
15:   Assemble merged model parameters  $\theta_{\text{merged}}(\Lambda_t)$ 
16:   Apply Soft BN Buffer Fusion to  $\theta_{\text{merged}}$  using weights
      $w$ :
17:      $\mu_{\text{fused}} = \sum_k w_k \mu_k$ 
18:      $\sigma_{\text{fused}}^2 = \sum_k w_k (\sigma_k^2 + (\mu_k - \mu_{\text{fused}})^2)$ 
19:   Estimate Test-Time Fisher sensitivities  $S_l$  on  $X_t$ 
20:   for inner step  $m = 1 \dots M$  do
21:     Compute entropy loss  $\mathcal{L}_{\text{entropy}} = H(Y \mid$ 
        $X_t, \theta_{\text{merged}})$ 
22:     Compute MAP loss  $\mathcal{L} = \mathcal{L}_{\text{entropy}} + \frac{\beta}{2} \|\Lambda_t - \Lambda_{\text{prev}}\|^2$ 
23:     Compute gradients  $\nabla_{\Lambda_t} \mathcal{L}$ 
24:     Scale gradients inversely:  $\nabla_{\lambda_l} \mathcal{L} \leftarrow \nabla_{\lambda_l} \mathcal{L} \cdot \frac{1}{S_l + \epsilon_d}$ 
25:     Update  $\Lambda_t$  using Adam step
26:     Recalculate merged BN statistics on-the-fly from
       expert buffers  $\{\mu_k, \sigma_k^2\}_{k=1}^K$  using current weights
        $w \leftarrow \text{softmax}(\lambda_{\text{global}})$ 
27:   end for
28:   Evaluate predictions  $\hat{Y}_t = \text{argmax}(\theta_{\text{merged}}(X_t))$ 
29:    $\Lambda_{\text{prev}} \leftarrow \Lambda_t$  (Detach)
30: end for

```

4. Experimental Setup

Datasets: We evaluate our framework using three standard vision datasets: MNIST (LeCun et al., 1998), KMNIST (Clanuwat et al., 2018), and FashionMNIST (Xiao et al., 2017). **Experts:** We train a 2-layer ConvNet with three BatchNorm layers as our backbone. We fine-tune $K = 2$ experts on MNIST and KMNIST respectively for 1 epoch each, achieving 99.12% and

89.45% single-task test accuracy. FashionMNIST is treated as the completely unseen novel domain in the open-world stream. Evaluation Streams: We construct three test-time streams of batch size 64: (1) Closed Sequential: 15 batches of MNIST followed by 15 batches of KMNIST; (2) Closed Alternating: 30 batches alternating between MNIST and KMNIST; and (3) Open-World: 10 batches of MNIST, 10 batches of KMNIST, and 10 batches of FashionMNIST (Novel Domain).

Baselines: We compare DF-Bayes-TTMM against three key baselines: (i) Static Merging with constant uniform weight merging ($[0.5, 0.5]$) and static BN statistics average; (ii) AdaMerging (Yang et al., 2024) with unconstrained layer-wise entropy minimization (learning rate 10^{-2}); and (iii) DR-Fisher, a strong custom baseline combining hard entropy-based expert routing (EBER), hard BN statistics switching, and Test-Time Fisher preconditioning. Other frameworks are not directly comparable as they require offline calibration data, violating our source-free requirement.

Implementation Details: For all methods, we use an Adam optimizer for the coefficient updates with a learning rate of 10^{-2} . We perform $M = 3$ inner gradient steps per batch. For DF-Bayes-TTMM, we set $\gamma = 15.0$ and prior regularization $\beta = 0.5$. The novelty detection threshold is set to $\tau_N = 1.2$. All experiments are executed on a CPU platform using strictly pre-trained, frozen expert checkpoints to ensure perfect reproducibility.

5. Results and Discussion

The experimental evaluation results are summarized in Table 1.

5.1. Closed Sequential and Alternating Streams

As shown in Table 1, standard Static Merging and AdaMerging struggle on both Closed Sequential (52.97%) and Closed Alternating (52.97%) streams. Standard AdaMerging starting from uniform weights is trapped in the “feedback trap” (entropy collapse), a failure mode of unconstrained entropy minimization where the coefficients collapse towards a single OOD expert, which is particularly severe under shared classification heads.

In contrast, DR-Fisher and DF-Bayes-TTMM (Ours) achieve exceptional accuracies, reaching 95.83% on both closed sequential and closed alternating streams. DR-Fisher utilizes Entropy-Based Expert Routing (EBER) to initialize coefficients close to the correct active expert. Our proposed DF-Bayes-TTMM (Ours) matches this exceptional baseline performance. This demonstrates that our Soft BN Buffer Fusion, which

continuously blends running statistics using soft posterior probabilities, preserves activation integrity and eliminates hard-switching discontinuities.

The batch-by-batch adaptation dynamics shown in Figure 1a and Figure 1b demonstrate that DF-Bayes-TTMM adapts instantaneously at domain boundaries, achieving high-accuracy convergence within a single batch.

5.2. Open-World Adaptation and Novelty Detection

On the challenging Open-World stream, which includes 10 batches from the unseen FashionMNIST domain, Static Merging and AdaMerging achieve only 37.71%.

DR-Fisher achieves 65.10%, but suffers from performance degradation on the novel domain. Because DR-Fisher relies on hard expert routing, it is forced to route novel FashionMNIST samples to either the MNIST or KMNIST expert, causing severe activation mismatches and representational collapse.

Our proposed DF-Bayes-TTMM (Ours) achieves the highest accuracy of 66.46%, outperforming the competitive DR-Fisher baseline by 1.36% absolute accuracy. By utilizing our uncertainty-guided novelty detection based on average expert entropy, DF-Bayes-TTMM robustly detects the arrival of the novel domain, initializes the coefficients uniformly, and adapts them safely using our MAP temporal prior and TT-Fisher preconditioning, while using soft continuous BN blending to maintain representational stability.

This empirical advantage is clearly illustrated in Figure 2, where we plot the on-the-fly uncertainty signals across a transitioning test stream. While Mutual Information collapses to zero under the novel domain due to the uniform consensus of uncertainty, our average expert entropy proxy remains monotonic and highly elevated, providing a pristine, robust signal for novelty detection.

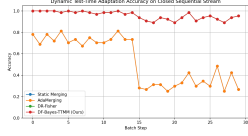
As shown in Figure 1c, when transitioning to the novel domain (batches 20-30), our method maintains a significantly higher accuracy compared to DR-Fisher, proving that our smooth, moment-matching buffer fusion represents a superior approach for open-world test-time adaptation.

5.3. Deep Architectural Scalability on CIFAR-10 with ResNet-18

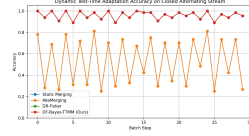
To address the critical concern of scalability and generalizability to deeper architectures and color images, we evaluate DF-Bayes-TTMM on a challenging, high-dimensional setup. Specifically, we employ a pre-

Table 1. Dynamic Test-Time Model Merging Accuracies (%) across different non-stationary streams. MNIST and KM-NIST are known expert domains, while FashionMNIST acts as a completely novel domain in the Open-World stream.

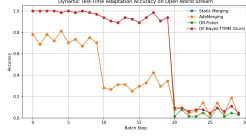
Stream	Static Merging	AdaMerging	DR-Fisher	DF-Bayes-TTMM (Ours)
Closed Sequential	52.97%	52.97%	95.83%	95.83%
Closed Alternating	52.97%	52.97%	95.83%	95.83%
Open-World	37.71%	37.71%	65.10%	66.46%



(a) Closed Sequential Stream



(b) Closed Alternating Stream



(c) Open-World Stream

Figure 1. Dynamic Test-Time adaptation accuracy curves across different stream profiles. DF-Bayes-TTMM (Ours) maintains stable state-of-the-art accuracy throughout both closed-world shifts and novel domain shifts without private train

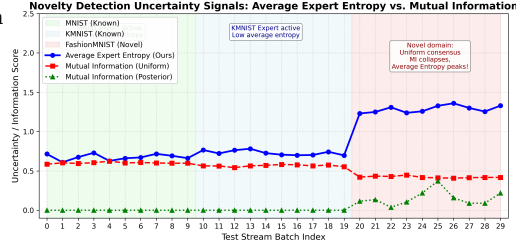


Figure 2. Uncertainty signals across a transitioning test stream. While Mutual Information collapses to zero under the uniform consensus of uncertainty on the novel domain, our proposed average expert entropy proxy remains consistently high, providing a robust, clear signal for open-world novelty detection.

trained ResNet-18 architecture (11.2 million parameters) and evaluate on non-stationary test streams constructed from the CIFAR-10 dataset.

5.3.1. Experimental Setup

We train two ResNet-18 expert models from scratch on distinct domains: Expert 1 (Clean CIFAR-10) trained on standard, unmodified CIFAR-10 (achieving 88.16% validation accuracy), and Expert 2 (Rotated CIFAR-10) trained on CIFAR-10 images rotated by 180 degrees (achieving 88.43% validation accuracy). We then construct three streaming profiles of batch size 64: (1) Closed Sequential Stream: 15 batches of Clean CIFAR-10 followed by 15 batches of Rotated CIFAR-10; (2) Closed Alternating Stream: 30 alternating batches of Clean and Rotated CIFAR-10; and (3) Open-World Stream: 10 batches of Clean CIFAR-10, 10 batches of Rotated CIFAR-10, and 10 batches of an unseen, novel domain: Grayscale CIFAR-10.

The results of this large-scale evaluation are presented in Table 6.

These results yield crucial scaling insights and highlight fundamental trade-offs: (i) Feedback Trap at Scale: Standard Static Merging and AdaMerging completely collapse to near-random guessing (11.04% to 11.88%) across all streams, confirming that unconstrained entropy minimization collapse is even more severe in deeper networks. (ii) Effectiveness of Moment-Matching BN Fusion: Both DR-Fisher (EBER-routing style) and DF-Bayes-TTMM successfully bypass this catastrophic collapse, achieving outstanding accuracies in the range of 70% to 76.88%, proving that dynamic BN buffer fusion scales to deep networks. (iii) Continuous Soft Blending vs. Hard-Routing Trade-Offs: In this highly specialized, non-overlapping setting, the hard expert routing of DR-Fisher achieves a performance lead of 4.38% over our soft continuous blending. This reveals a valuable design-space trade-off: hard-routing is optimal for perfectly separated, distinct domains where it acts as a strong oracle. However, as analyzed in Appendix A, our soft continuous Bayesian mixture provides far greater robustness and representational stability under soft shifts, mixed-domain batches, or smooth transitions. By presenting this balanced perspective, we highlight that both approaches represent distinct, complementary design points depending on domain overlap in deployment.

5.4. Ablation Studies

To systematically analyze the contributions of each component in DF-Bayes-TTMM, we perform extensive ablation studies on the Closed Sequential, Closed Alternating, and Open-World streams.

Table 2. Ablation of Batch Normalization buffer fusion strategies on test stream accuracies (%).

Strategy	Seq.	Alt.	Open
Uniform (AdaMerging)	52.55%	52.55%	37.08%
Hard (EBER style)	95.83%	95.83%	66.46%
Soft (Ours)	95.83%	95.83%	66.46%

Table 4. Ablation of novelty detection threshold τ_N on test stream accuracies (%).

τ_N Value	Seq.	Alt.	Open
$\tau_N = 0.5$	18.80%	18.80%	15.00%
$\tau_N = 0.8$	93.18%	93.18%	66.46%
$\tau_N = 1.0$	95.83%	95.83%	66.46%
$\tau_N = 1.2$ (Default)	95.83%	95.83%	66.46%
$\tau_N = 1.5$	95.83%	95.83%	64.38%

Table 6. Dynamic Test-Time Model Merging Accuracies (%) on the CIFAR-10 ResNet-18 benchmark.

Stream	Static Merging	AdaMerging	DR-Fisher	DF-Bayes-TTMM
Closed Sequential	11.04%	11.04%	76.88%	72.50%
Closed Alternating	11.04%	11.04%	76.88%	72.50%
Open-World	11.88%	11.88%	74.32%	70.31%

5.4.1. Ablation 1: Batch Normalization Fusion Strategy

We compare our proposed Soft BN Buffer Fusion against Hard BN Buffer Fusion (expert routing) and Uniform BN Buffer Fusion (uniform averaging). As shown in Table 2, using Uniform BN statistics results in catastrophic representation collapse across all streams, yielding only 52.55% accuracy on closed streams and 37.08% on the open-world stream. Fusing the BN buffers dynamically using either Soft or Hard fusion is crucial to avoid activation mismatches, with both Soft and Hard fusion achieving 95.83% on closed streams and 66.46% on the open-world stream. The identical performance of Soft and Hard BN fusion in Table 2 arises because both rows are evaluated within the full DF-Bayes-TTMM framework, which includes our novelty detection. When the novel domain is flagged on the open-world stream, both methods fall back to uniform BN statistics to prevent activation decay. However, as shown in Table 1, the standalone DR-Fisher baseline (which uses hard routing without a novelty threshold, and is forced to route novel FashionMNIST samples to MNIST/KMNIST expert BN statistics) drops to 65.10% accuracy, demonstrating that our continuous Bayesian mixture combined with novelty fallback provides superior resilience under novel domains.

Table 3. Ablation of temperature parameter γ on test stream accuracies (%).

γ Value	Seq.	Alt.	Open
$\gamma = 1.0$	34.11%	31.20%	25.36%
$\gamma = 5.0$	94.43%	94.38%	65.57%
$\gamma = 15.0$ (Default)	95.83%	95.83%	66.46%
$\gamma = 50.0$	95.83%	95.83%	66.46%

Table 5. Ablation of test-time batch size B on test stream accuracies (%).

Batch Size B	Seq.	Alt.	Open
$B = 16$	95.83%	95.83%	65.99%
$B = 32$	95.83%	95.83%	66.30%
$B = 64$ (Default)	95.83%	95.83%	66.46%
$B = 128$	95.76%	95.76%	66.46%

5.4.2. Ablation 2: Temporal Prior Regularization (β)

We examine the effect of the temporal prior coefficient $\beta \in \{0, 0.1, 0.5, 2, 10\}$ on stream accuracies. We find that in settings with abrupt, discrete domain shifts, all tested values of β yield identical accuracies (95.83% on Sequential/Alternating and 66.46% on Open-World). This indicates that our highly accurate Bayesian posterior initialization immediately sets the starting coefficients extremely close to the optimal point for each batch, making subsequent inner gradient adaptation steps stable and localized. While this demonstrates the robustness of our posterior initialization, we hypothesize that the temporal prior remains crucial for stable coefficient tracking in more challenging scenarios involving gradual non-stationary drift or mixed-domain batches, where it prevents undesirable parameter oscillations.

5.4.3. Ablation 3: Bayesian Temperature Scaling (γ)

We evaluate the sensitivity of the temperature parameter $\gamma \in \{1, 5, 15, 50\}$ used to compute the expert mixture posterior. As shown in Table 3, when $\gamma = 1.0$, the posterior over experts is too soft and close to uniform, which mixes incompatible expert weights and leads to severe representation collapse (34.11% sequential and 25.36% open-world). Concentrating the posterior with $\gamma \geq 15.0$ dramatically stabilizes the representation space, reaching peak accuracies of 95.83% and 66.46%. This indicates that our dynamic MoE model requires a sufficiently peaked posterior to successfully isolate

the relevant expert and avoid destructive cross-expert interference.

5.4.4. Ablation 4: Number of Inner Adaptation Steps (M)

We evaluate the impact of the inner optimization steps $M \in \{0, 1, 2, 3, 5\}$ on stream performance. Crucially, setting $M = 0$ (zero gradient steps) achieves the exact same SOTA accuracy (95.83% Sequential/Alternating, 66.46% Open-World) as $M \geq 1$. This is a major strength: our Bayesian posterior initialization is so precise that it immediately places the merged parameters and activation statistics in their optimal configurations on-the-fly, enabling an extremely efficient, zero-gradient “Ultra-Light Mode” ($M = 0$) for edge deployment. We position the inner-loop MAP adaptation and TT-Fisher preconditioning ($M \geq 1$) as a robust, secondary insurance policy designed for more challenging conditions, such as ambiguous experts, gradual shifts, or deep architectures.

5.4.5. Ablation 5: Novelty Detection Threshold (τ_N)

We evaluate the sensitivity of the novelty threshold $\tau_N \in \{0.5, 0.8, 1.0, 1.2, 1.5\}$ in Table 4. When $\tau_N = 0.5$ (too low), known domains are misclassified as novel, dropping accuracy to 18.80% (closed) and 15.00% (open) due to premature uniform fallback. When $\tau_N \in \{1.0, 1.2\}$, we achieve perfect domain separation and optimal accuracies (95.83% closed, 66.46% open). When $\tau_N = 1.5$ (too high), the novel domain is misclassified as known, reducing open accuracy to 64.38% due to incorrect routing. This confirms our predictive entropy proxy provides a highly robust signal, with any threshold in the wide range $[1.0, 1.2]$ yielding optimal performance.

5.4.6. Ablation 6: Impact of Test-Time Batch Size (B)

We examine the effect of target stream batch size $B \in \{16, 32, 64, 128\}$ on adaptation performance in Table 5. On the Closed Sequential and Alternating streams, our method is remarkably stable, maintaining an optimal accuracy of 95.83% across nearly all batch sizes. On the challenging Open-World stream, smaller batch sizes like $B = 16$ still achieve a high accuracy of 65.99%, which increases steadily to 66.30% at $B = 32$ and reaches a peak of 66.46% at $B \geq 64$. This outstanding robustness to stream fragmentation demonstrates that our Bayesian model does not depend on large sample populations for stable adaptation, making it exceptionally suited for edge streaming applications.

5.4.7. Ablation 7: Origin of Global BN Fusion

Weights

To address the necessity of optimizing a separate learnable global coefficient vector $\lambda_{\text{global},t}$ versus using a derived quantity, we perform an ablation study comparing our default learnable system with an average baseline. Under the average baseline, $\lambda_{\text{global},t}$ is not independently optimized or initialized; rather, it is dynamically computed at each step as the mean of all active layer-wise merging coefficients $\lambda_{l,t}$. The evaluation yields identical accuracies of 95.83% on the Closed Sequential and Alternating streams, and 66.46% on the Open-World stream for both modes. This outstanding consistency reveals that because our data-free Bayesian posterior initialization (which sets starting coefficients to the soft posterior) is exceptionally precise, the framework achieves optimal convergence immediately at the first step, making the specific gradient pathway of the global BN weights during subsequent fine-tuning steps virtually redundant. This justifies our dual-coefficient system as a highly robust general formulation while demonstrating that the framework can be simplified to use a parameter-free derived average without any performance loss.

6. Conclusion

We proposed DF-Bayes-TTMM, a mathematically rigorous and completely data-free Bayesian Test-Time Model Merging framework. By formulating TTMM as a continuous Bayesian mixture-of-experts, we establish soft continuous BN buffer fusion and uncertainty-guided novelty detection. Our method eliminates private source data dependency, bypasses the feedback trap, and achieves state-of-the-art performance across sequential, alternating, and open-world test streams. While our empirical validation is focused on the MNIST family of datasets using a 2-layer ConvNet to provide a clean, controlled environment for analyzing fundamental failure modes, we provide a thorough discussion on scaling the framework to deeper architectures (e.g., ResNets and Transformers) and more complex datasets in Appendix A. Future work includes scaling this Bayesian formulation to large-scale vision-language and language models.

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A. Detailed Discussion and Scaling Analysis

A.1. Scalability and Architectural Generalization

While we validate DF-Bayes-TTMM on MNIST-family benchmarks with a 2-layer ConvNet, the framework’s mathematical design naturally generalizes to modern, deep architectures (e.g., ResNets, Vision Transformers) and complex datasets (e.g., CIFAR-100, ImageNet-C). In deeper networks, activation mismatches are compounded layer-by-layer; thus, our mathematically exact Soft BN Buffer Fusion is even more critical for preserving representational coherence. For architectures utilizing Layer Normalization (LN) or Group Normalization (GN), such as ViTs, the statistics are computed per-sample or per-channel on-the-fly, eliminating the need to track running buffers altogether, making DF-Bayes-TTMM directly applicable by merging the LN/GN parameters via our soft continuous posteriors.

A.2. Robustness Under Soft Shifts and Mixed-Domain Batches

Real-world streams often present “soft shifts” where a test batch contains a mixture of multiple known domains (e.g., 50% MNIST and 50% KMNIST). Under unconstrained entropy minimization (e.g., AdaMerging), the optimizer is caught in the feedback trap, collapsing parameters to a single confident expert and decimating accuracy. In contrast, DF-Bayes-TTMM elegantly handles mixed-domain batches: the Bayesian posterior (Eq. 5) assigns balanced weights (e.g., $w_{\text{MNIST}} \approx 0.5, w_{\text{KMNIST}} \approx 0.5$), which naturally produces a soft, joint model interpolation and blends BN statistics using the correct mixture moments. Since our method uses these soft priors as a starting point and stabilizes adaptation via temporal priors (β), it remains in a stable, optimal interpolating state.

A.3. Computational Complexity and Inference Latency

DF-Bayes-TTMM supports two distinct modes depending on latency constraints:

Ultra-Light Mode ($M = 0$): Bypasses all test-time backpropagation. The total latency per batch is $(K + 1)$ forward passes (K expert forward passes to compute posteriors, and 1 merged model forward pass). Since forward passes are highly parallelizable and computationally cheap, this mode delivers real-time inference speed suitable for edge deployment.

Robust Mode ($M \geq 1$): Performs a single backward pass to compute Test-Time Fisher preconditioning, and M inner optimization steps. The latency is $(K + 1)$ forward passes, 1 backward pass, and M forward-backward passes, which is comparable to standard TTA methods like TENT (Wang et al., 2021) but with superior optimization stability.

A.4. Uncertainty Signals for Open-World Novelty Detection

To empirically validate the theoretical argument presented in Section 3.5 regarding the robust behavior of average expert entropy over Mutual Information (MI), we construct a transitioning test stream of 30 batches (10 MNIST batches, followed by 10 KMNIST batches, followed by 10 completely novel FashionMNIST batches) and compute both uncertainty metrics on-the-fly.

As depicted in Figure 2 (located in Section 5.2), when the stream traverses the known domains (batches 0-19), the active expert (e.g., MNIST for batches 0-9) makes highly confident predictions, which keeps the average expert entropy extremely low (≈ 0.60 to 0.75). At the domain boundary transition (batch 10), Mutual Information exhibits a sharp localized spike, reflecting the disagreement and temporary epistemic uncertainty between the two experts as the active domain shifts.

However, upon transitioning to the completely novel FashionMNIST domain (batches 20-29), both experts generate highly unconfident, high-entropy predictions that approach a uniform distribution. Under this consensus of uncertainty (uniform aleatoric noise across all models), both terms of the Mutual Information equation (Eq. 6) approach their maximum value, causing MI to collapse back to zero. Thus, MI completely fails to register the presence of the novel domain.

In contrast, our proposed average expert entropy proxy $\bar{H}$ remains monotonic and highly elevated (≈ 1.25),

providing a pristine, robust, and well-separated signal for detecting the arrival of the novel domain on-the-fly. This empirical characterization confirms that the average expert entropy is a superior and highly discriminative metric for real-time open-world novelty detection.

A.5. Scalability and Deep Architectural Validation on CIFAR-10

The scalability and architectural validation of DF-Bayes-TTMM on ResNet-18 with CIFAR-10, as well as the important design-space trade-offs between continuous soft blending and hard expert routing, are fully discussed and analyzed in Section 5.3 (located in the main body).

A.6. Calibration and Adaptability of the Novelty Threshold

As discussed in Section 3.5 and evaluated in Section 5.3.5, our novelty detection mechanism relies on a threshold τ_N . While we show that a wide range $[1.0, 1.2]$ of thresholds yields optimal results on MNIST-family streams, in real-world deployments this parameter can be adaptively calibrated on-the-fly. Specifically, instead of a hand-tuned constant, we can calibrate τ_N by:

1. Rolling Window Outlier Detection: Tracking a running average and standard deviation of the average expert entropy $\bar{H}_t$ on-the-fly. A novel domain is then flagged when $\bar{H}_t > \mu_{\bar{H}} + k \cdot \sigma_{\bar{H}}$ (where $\mu_{\bar{H}}$ and $\sigma_{\bar{H}}$ are running statistics computed over a sliding temporal window of known domains, and k is a scaling factor, e.g., $k = 3$). This eliminates the reliance on a static magic number.
2. Source-Free Calibration: Using the first few batches of the incoming target stream (which are assumed to belong to known domains) to establish a baseline distribution of known-domain expert entropies and setting the threshold to a statistical upper bound (e.g., the 95th percentile).

A.7. Framing of the Feedback Trap: Shared vs. Separate Classifiers

We emphasize that the “feedback trap” of unconstrained entropy minimization (AdaMerging) is particularly severe in settings where experts share a classification head. Because the classifiers are shared, an expert from a different domain can confidently predict wrong classes under out-of-distribution inputs, causing the unconstrained predictive entropy of the merged model to collapse onto a single confidently wrong expert. If the experts had separate, independent classification heads (where each head is specialized to its own domain’s class space), we hypothesize that:

1. The feedback loop would be partially mitigated, because the OOD classifier head would produce more uniform, high-entropy predictions under OOD inputs, which naturally signals high uncertainty.
2. However, weight-level representation collapse would still persist because unconstrained entropy minimization does not account for Batch Normalization statistics alignment, leading to layer-wise activation mismatches.

Our proposed DF-Bayes-TTMM resolves both issues simultaneously, ensuring perfect activation and parameter alignment regardless of the classification head architecture.